



Healing the Planet and Patients: Evaluating a New Climate and Health Curriculum

Abstract

In 2023, Harvard Medical School began weaving lessons on climate change, environment, and health into its existing courses. A 17-month evaluation published in PLOS Climate in 2025 found that first-year medical students greatly improved their knowledge of climate-related health and equity topics. This lesson explores how researchers measured the curriculum's impact and what it means for training future doctors.

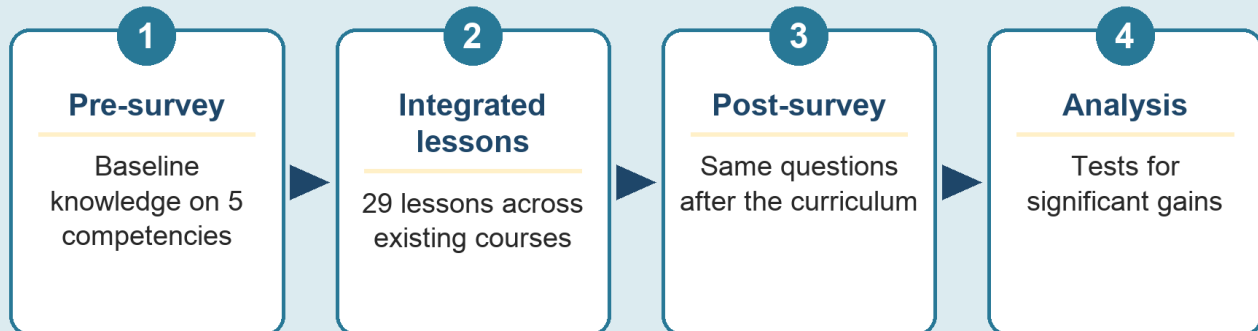
Introduction

Climate change introduces new risks to patients, from extreme heat affecting certain medications to wildfire smoke worsening lung disease. Future doctors will be on the front lines of these challenges, but many medical schools offer only standalone lectures on climate and health, and there is no standard approach. Harvard Medical School took a different path. Launched in fall 2023, its climate change, environment, and health (CCEH) curriculum embeds lessons into existing courses. Students learn how climate change affects health, health equity, and health care delivery; how the health system itself contributes to climate change; and how health professionals and institutions can be part of the solution.

How Do Scientists Study This?

Researchers from Harvard Medical School and the Harvard T.H. Chan School of Public Health used pre- and post-surveys to measure what first-year students learned. At the start and end of the curriculum, students rated their knowledge on five core competencies, using 20 survey items. The team also surveyed students before and after each of 29 lesson integrations, including lectures, case-based learning sessions, small group discussions, and preparatory materials, to see how each one affected understanding and confidence. They then used statistical tests to determine which gains were significant, meaning unlikely to be due to chance, and measured effect sizes to judge how large the changes were.

How the curriculum was evaluated



Survey before

Students rated their knowledge of five core climate and health competencies before the curriculum.

Teach in every course

Climate topics were integrated into 29 lessons across existing pre-clerkship courses.

Survey after

Students rated their knowledge again, and statistical tests measured the change.

What Did They Discover?

Students reported large knowledge gains across all five core competencies, and 19 of 20 survey items showed statistically significant improvement. The share of students who could define how climate change, air pollution, and ecological damage harm health rose from 33% to 92%. The share able to apply that knowledge to patient care jumped from 19% to 92%. Understanding of how the health care system contributes to climate change rose from 41% to 96%. All 29 lesson integrations produced gains, and 20 were statistically significant. The largest gains included taking an environmental history from a patient and understanding environmental racism. Some clinical skills, like counseling patients about air pollution, improved only moderately, suggesting a need for more hands-on training.

Student knowledge gains

19→92% could apply climate-health knowledge to patient care

41→96% understood health care's role in climate change

87% of students found the curriculum valuable

87% of students who completed the post-survey found the curriculum valuable, and effect sizes were large across most survey items.

Takeaways

Integration works

Weaving climate into existing courses produced large, significant knowledge gains.

Measure to improve

Pre- and post-surveys showed which lessons worked and where more practice is needed.

A model for others

Other medical schools can use this approach to prepare doctors for climate change.

Key Terms

Curriculum integration

Embedding a topic throughout existing courses instead of teaching it only in separate lessons.

Core competency

A key skill or area of knowledge that learners are expected to master.

Pre- and post-survey

Questionnaires given before and after a program to measure how much participants changed.

Statistical significance

A result unlikely to have happened by chance, based on a statistical test.

Environmental racism

The unequal exposure of communities of color to pollution and environmental hazards because of policies and practices.

Words to Know · Healing the Planet and Patients

Curriculum integration

Core competency

Pre- and post-survey

Statistical significance

Environmental racism

Activity

Measure Your Own Learning

Students run a mini pre- and post-survey on this lesson to experience how curricula are evaluated.

Materials: Pre-survey and post-survey handouts with 1 to 5 rating scales; Calculator; Graph paper or spreadsheet

1. Before reading the findings, rate your confidence on three statements, such as 'I can explain how heat affects patients.'
2. Read the findings section and discuss it with a partner.
3. Rate the same three statements again.
4. As a class, calculate the average before and after for each statement and graph the change.
5. Discuss: what are the limits of self-reported knowledge? How else could you measure learning?

References

Miller, A. (2025). Students report major success of new climate and health curriculum at Harvard Medical School. Harvard Chan C-CHANGE.
<https://hsph.harvard.edu/climate-health-c-change/news/students-report-major-success-of-new-climate-and-health-curriculum-at-harvard-medical-school/>

Baker, N., Shirley, H., et al. (2025). Evaluating the impact of a longitudinal, integrated climate change, health, and environment curriculum in undergraduate medical training at Harvard Medical School. PLOS Climate, 4(12), e0000727. <https://doi.org/10.1371/journal.pclm.0000727>